

Fourier Transforms

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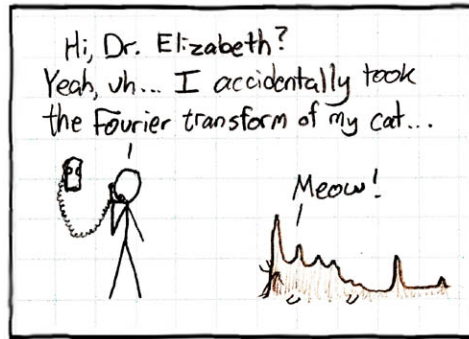


Figure 1: Fourier. Source: Randall Munroe, <http://www.xkcd.org>.

1 Continuous Fourier Transform

The **continuous Fourier transform** (FT) was first published by Joseph Fourier in 1822 and is used to decompose a function into periodic functions. FT's find a wide array of uses from identifying the composition of stars (spectroscopy) to cell phone transmissions to image processing to solving differential equations.

Sometimes shorthand notation for a FT is given by

$$h(\xi) \xleftrightarrow{\mathcal{F}} H(\nu), \quad (1)$$

where $h(\xi)$ is a function with the FT $H(\nu)$. **The forward transform** goes from left to right and **inverse transform** from right to left.

Forward transform

$$\mathcal{F}[h(\xi)] = H(\nu) = \int_{-\infty}^{\infty} h(\xi) \exp[-i2\pi(\nu \cdot \xi)] d\xi, \quad (2)$$

Inverse transform

$$\mathcal{F}^{-1}[H(\nu)] = h(\xi) = \int_{-\infty}^{\infty} H(\nu) \exp[i2\pi(\nu \cdot \xi)] d\nu. \quad (3)$$

The complex number $i = \sqrt{-1}$ (Figure 2). Note that j is occasionally used as the complex number, particularly in electrical engineering texts. Recall that $\exp(i\theta) = \cos(\theta) + i\sin(\theta)$, hence, a FT and its inverse can be represented by a series of trigonometric functions with varying amplitudes and frequencies.

$$\exp(i2\pi ft) = \cos(2\pi ft) + i\sin(2\pi ft) \text{ and} \quad (4)$$

$$\exp(-i2\pi ft) = \cos(2\pi ft) - i\sin(2\pi ft). \quad (5)$$

FT's can be performed on temporal (i.e. **time-domain**) or spatial (i.e. **spatial-domain**) processes converting a function into the **frequency-domain**.

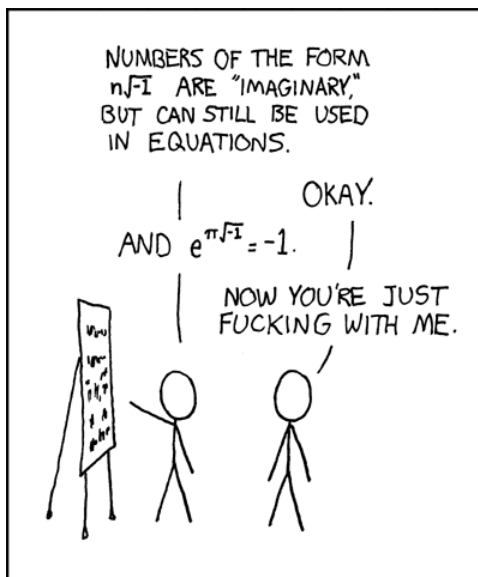


Figure 2: e to the π times i . Source: Randall Munroe, <http://www.xkcd.org>.

1.1 Time-domain

- time: $\xi = t$
- frequency: $\nu = f$ or angular frequency: $2\pi f = \omega$
- period: $f^{-1} = \tau$
- forward:

$$H(f) = \int_{-\infty}^{\infty} h(t) \exp[-i2\pi ft] dt, \quad (6)$$

- inverse:

$$h(t) = \int_{-\infty}^{\infty} H(f) \exp[i2\pi ft] df. \quad (7)$$

1.2 Spatial-domain

- space: $\xi = \mathbf{x} = \{x_0, \dots, x_{N-1}\}$
- wavenumber: $\nu = \mathbf{k} = \{k_{x_0}, \dots, k_{x_{N-1}}\}$
- wavelength: $k^{-1} = \lambda$
- forward:

$$H(\mathbf{k}) = \int \cdots \int_{-\infty}^{\infty} h(\mathbf{x}) \exp[-i2\pi(\mathbf{k} \cdot \mathbf{x})] d^{N-1}\mathbf{x}, \quad (8)$$

- inverse:

$$h(\mathbf{x}) = \int \cdots \int_{-\infty}^{\infty} H(\mathbf{k}) \exp[i2\pi(\mathbf{k} \cdot \mathbf{x})] d^{N-1}\mathbf{k}. \quad (9)$$

Hence for two dimensions, the FT is written

$$H(k_x, k_y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) \exp[-i2\pi(k_x x + k_y y)] dx dy, \quad (10)$$

1.3 Amplitude and Phase

A forward FT'ed function is complex and therefore can be written as,

$$H(\nu) = A(\nu) + iB(\nu) = |H(\nu)| \exp[i\phi(\nu)], \quad (11)$$

where $A(\nu)$ and $B(\nu)$ are the real and imaginary parts of $H(\nu)$, respectively. We often refer to the **amplitude spectrum**, $|H(\nu)|$, and the **phase spectrum**, $\phi(\nu)$ of a signal. We can write these two spectra in terms of the real and imaginary parts of $H(\nu)$,

$$|H(\nu)| = [A^2(\nu) + B^2(\nu)]^{1/2}, \text{ and} \quad (12)$$

$$\phi(\nu) = \tan^{-1} \left[\frac{B(\nu)}{A(\nu)} \right]. \quad (13)$$

2 Discrete Fourier Transform (DFT)

The DFT was first developed, although never published, by Heinrich Gauss in 1805 to describe the orbits of asteroids using discrete observations. It was not until Cooley and Turkey in 1965 that the DFT was re-developed and is equivalent to Gauss's solution. In most sciences, including geophysics, the DFT is more useful than the CFT in most cases since our data are made as a series of discrete observations and limited in length or duration. The DFT prevents issues with integration from $-\infty$ to ∞ by using finite sums and uses discrete values rather than continuous functions.

For a discrete set of data, $\mathbf{d} = \{d_0, d_1, \dots, d_{N-1}\}$, the DFT is given by an identical length vector, $\mathbf{D} = \{D_0, D_1, \dots, D_{N-1}\}$, i.e. $\mathbf{d} \xrightarrow{\mathcal{F}} \mathbf{D}$.

Forward discrete transform

$$D_k = \sum_{n=0}^{N-1} d_n \exp \left[-\frac{i2\pi}{N} kn \right], \text{ where } k = 0, \dots, N-1, \quad (14)$$

Inverse discrete transform

$$d_n = \frac{1}{N} \sum_{k=0}^{N-1} D_k \exp \left[\frac{i2\pi}{N} kn \right], \text{ where } n = 0, \dots, N-1. \quad (15)$$

Note that within MatLab, indices into arrays must start at 1, not 0. Hence the sums are over 1 to N with $n \rightarrow n-1$ and $k \rightarrow k-1$.

Equations 14 and 15 assume that the sample interval is unit length. This is all well and good, except that we wish to relate the discrete frequencies to the real-world frequencies. The discrete frequencies k are related to the sampling interval, $\Delta\xi$ by

$$\nu_k = [(k-1)\Delta\xi]^{-1}, \text{ where } k = 0, \dots, N-1. \quad (16)$$

You may recall that $\nu_s = (2\Delta\xi)^{-1}$ is the Nyquist (sampling) frequency.

3 Aliasing

The sampling rate (digitize) of a continuous continuous signal controls the maximum frequency content that is represented by the digitized signal. Theoretically, the highest frequency that can be represented by a digital signal is the Nyquist frequency, $f_{Ny} = (2\Delta t)^{-1}$. When a signal is under-sampled, aliased, the signal is distorted and often not in an obvious and detectable way. The term aliasing comes from the Latin word *alias*, meaning “otherwise or at another time” and similarly in English as a “false name typically for the purposes of deception”. In much the same way aliasing a signal does a the same to the frequencies below the Nyquist frequency.

For example, have you ever watched the tires on a car that is accelerating? Watch the wheels spin forward at an increasing rate. At a particular speed, they appear to stop and then rotate backwards. This phenomena occurs as a result of your brain aliasing the visual signal. The same effect can also be seen in rotating helicopter blades or air circulating fans.

Effects of Aliasing

The effects of aliasing are severe. There are four effects:

1. “*You can’t always get what you want.*” The original signal (for frequencies higher than Nyquist) is not correctly represented or even contained in the sampled data.
2. “*You get something else instead.*” The original high frequency signal aliases as a lower frequency signal that is spurious (is not included in the original data).

3. *“You don’t know that you’ve made a mistake.”* Once you have sampled, you don’t know that the sampled data are incorrect—the data look like perfectly reasonable data.
4. *“It’s too late! You can’t correct the mistake.”* Once you have sampled the data incorrectly (Δt too large), you can’t fix the problem.

Preventing Aliasing

1. Experimental design - Be aware of the frequency range you will encounter and sample at a rate above the Nyquist frequency.
2. Filtering - If sampling above the Nyquist frequency is not possible or too expensive (think data storage and processing), prefilter the signal before it is digitized (using electronics).